# Gated

User Manual

Michael Dungan

## What is Gated?

Gated chops incoming audio using a repeating rhythmic pattern. Eight slots hold on/off steps and ties, shaped by a shared ADSR envelope. Swing changes the groove; random population creates new rhythms without changing your sound controls.

## Installation

Build VST3, CLAP, and standalone with:

```
cargo xtask bundle -p gated --native
```

Bundles are written under `target/bundled/`. Copy the VST3 or CLAP bundle into your DAW's plugin directory. For GUI-only evaluation without an audio device:

```
cargo run -p gated -- --backend dummy
```

## Quick start

1. Insert Gated on sustained audio and start the DAW transport.
2. Click ring segments to turn steps off or on. All 16 default steps start on.
3. Drag across segments to enable and tie a longer held gate.
4. Adjust Gate and Release for choppy or flowing articulation. Add Swing.
5. Select another numbered slot to edit and hear another rhythm, or use Randomize.

## Controls

| Control | Range / default | Behavior |
|---|---|---|
| Pattern | 1–8 / 1 | Selects both the playing and edited slot; automatable. |
| Steps | 1–64 / 16 | Length of this slot. Use the central −/+ buttons. |
| Attack | 0–2000 ms / 2 ms | Time from zero to full level. |
| Decay | 0–2000 ms / 50 ms | Time from full level to Sustain. |
| Sustain | 0–100% / 100% | Held level after Attack and Decay. |
| Release | 0–4000 ms / 20 ms | Time from the note-off level to silence. |
| Resolution | 1/1–1/64 / 1/16 | Step duration; D = dotted, T = triplet. |

| Control | Range / default | Behavior |
|---|---|---|
| Gate | 1–100% / 50% | Portion of the final step held before Release. |
| Swing | 0–100% / 0% | Equal step pairs through a 3:1 long/short groove. |
| Mix | 0–100% / 100% | Linear blend of dry audio and gated audio. |
| Density | 0–100% / 50% | Chance that Randomize turns each step on. |
| Tie probability | 0–100% / 25% | Chance of a tie between adjacent on steps. |
| Bypass | Off | Passes input through while the clock follows the song. |

The steps, ties, and length belong to each slot. All other controls are global. Each slot initially contains the same 16 active, untied steps.

## Pattern editing

Click a segment to toggle it. Drag clockwise or counterclockwise to turn on and tie a run of steps. Hold Shift while dragging to paint steps off. Right-click an active step to toggle its tie to the next step; both steps must be on. A tie from the last step to the first continues across the loop boundary.

Untied on steps restart the envelope. Tied steps hold it, with Gate shortening only the last step of the run. An entirely on, entirely tied pattern holds at Sustain. Shortening a pattern retains hidden step data for later expansion.

The Copy/Paste shortcuts transfer a slot's complete pattern and length. The arrow buttons rotate the visible steps and ties one step left/right. Randomize changes only the current slot's visible steps and ties. It preserves length, hidden steps, other slots, and global controls. Density zero intentionally produces silence.

The Undo/Redo shortcuts operate on pattern changes. A drag or randomization is one undo action. Pattern history and the internal clipboard last while the editor is open; recalling externally saved pattern state clears the history.

## Timing and swing

Step one is anchored to musical beat zero. Starting playback in the middle of the song, seeking, and looping select the corresponding pattern position. Changing slots switches immediately at the incoming pattern's current song position; it does not restart the rhythm.

Swing delays the second step of each pair without changing the pair's total duration. Pairing starts again at each pattern cycle. For odd lengths, the last unpaired step stays straight. Swing applies to dotted and triplet resolutions too.

When transport stops, input passes through. A short gain slew avoids abrupt changes during stops, bypass, pattern switches, and envelope retriggers. If the host omits song position, Gated uses a tempo clock restarting on Play and displays that limitation. A missing tempo falls back to 120 BPM.

## Mouse and keyboard

- Drag a dial vertically; hold Shift for fine adjustment.

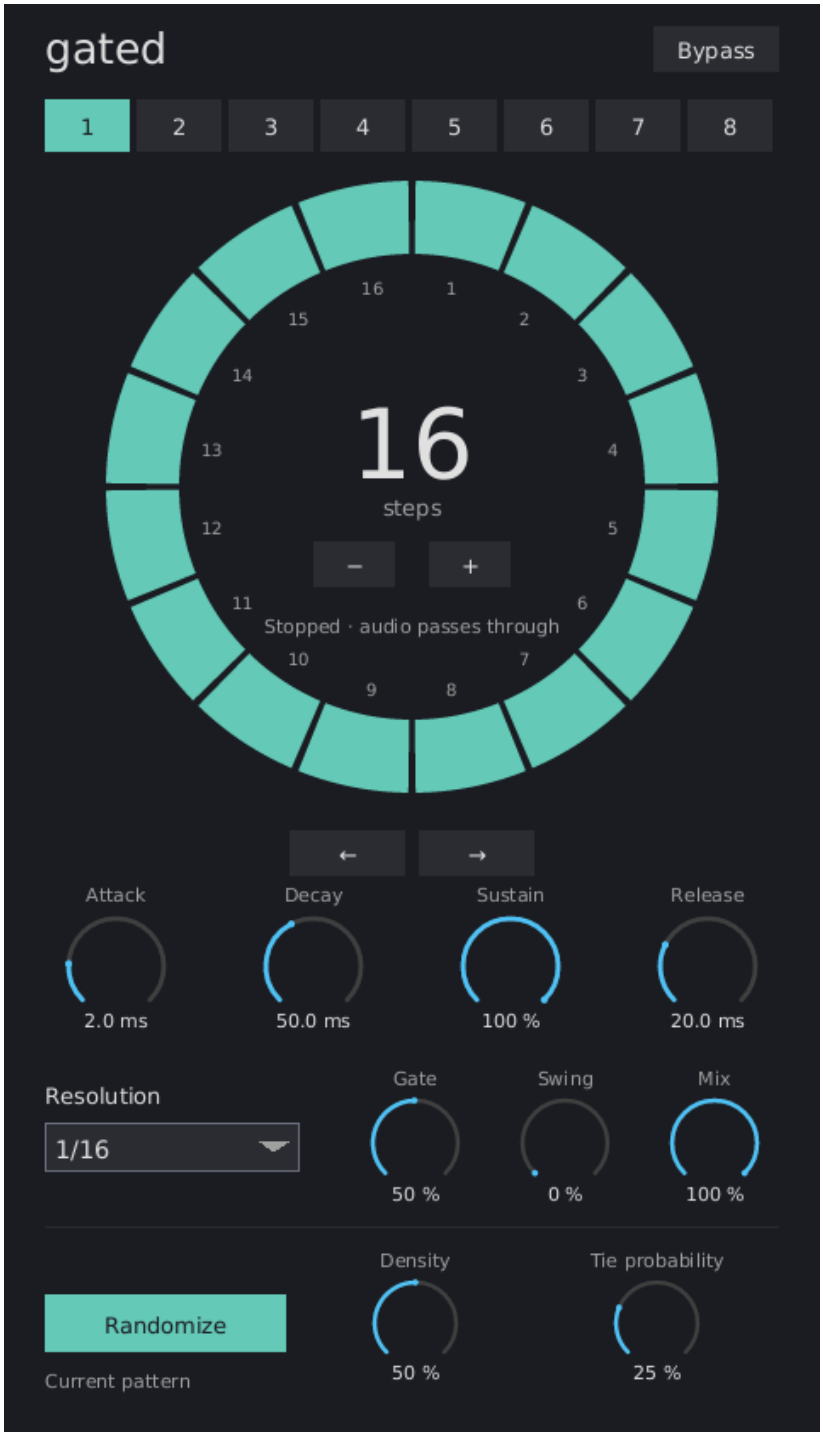



Figure 1: Gated interface

- Double-click a dial to restore its default.
- Right-click a dial to type a value; Enter commits, Escape cancels.
- Ctrl/Cmd+C and Ctrl/Cmd+V copy/paste patterns.
- Ctrl/Cmd+Z undoes; Ctrl/Cmd+Shift+Z or Ctrl/Cmd+Y redoes.
- The resolution menu supports arrows, Page Up/Down, Enter, Escape, and scrolling.

## Technical notes

Mono and stereo are supported. In stereo, the same gain applies to both channels. There is no lookahead or FFT and no reported latency. The editor renders on the CPU and scales to fit its freely resized window. Audio processing uses fixed storage and a nonblocking copy of pattern state.

Version one exposes 64 steps. A later 128-step limit and Euclidean population are planned separately.

## Formats and license

VST3, CLAP, and standalone. GPL-3.0-or-later, matching the plugin pack.